

Module-05

STORAGE, SECURITY AND MODERN CONCEPTS

Prof. Sadaf Malik

Assistant Professor, Computer Engg.

A.I.K.T.C New Panve

5

Storage, Security and Modern Concepts

5.1

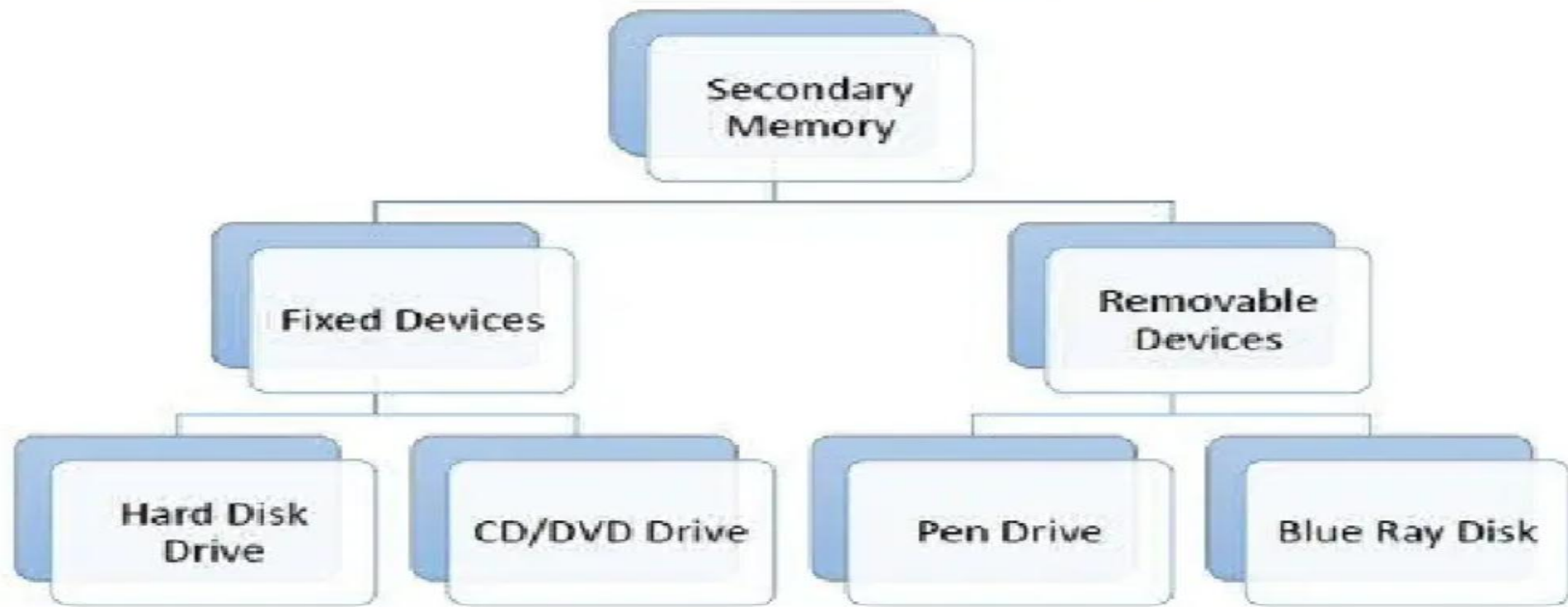
Secondary Storage Management
Disk Structure and Scheduling: FCFS, SSTF, SCAN,
C-SCAN

5.3

System Security and Protection
Goals of Protection, Access Matrix, Authentication

Secondary storage management

Secondary storage management by an operating system (OS) involves handling non-volatile storage devices like hard disk drives (HDDs) and solid-state drives (SSDs) to store programs, user data, and virtual memory. Key OS activities include managing free space, allocating storage to files, and scheduling I/O requests to optimize performance using algorithms like **FCFS**, **SSTF**, **SCAN**, and **LOOK**. The OS provides the **file system** interface for users to interact with these storage devices for long-term data persistence.



5.1 DISK STRUCTURE

What is Hard Disk ?

Hard Disk is a secondary storage device. It is a type of [electro mechanical device](#). A hard disk stores and retrieves digital data using magnetic storage. Disk is rigid rapidly rotating platters coated with magnetic material.

Hard Disk Pack consist of more than one disk. How Data is Stored in a Hard Disk ?

- Data is stored in Disk in form of tracks and sectors.
- Disk surface is divided in to tracks.
- Track is further dived in to sectors.
- Sector is the addressable unit in disk.

Basic picture for disk storage concept is given below

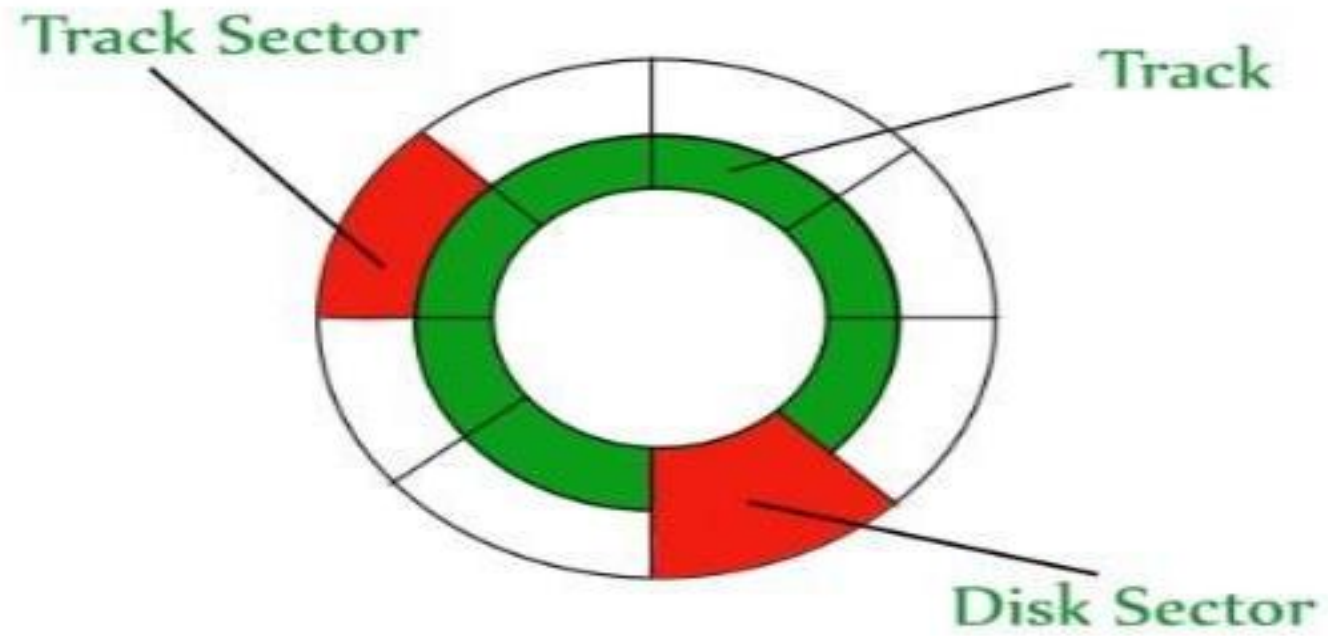
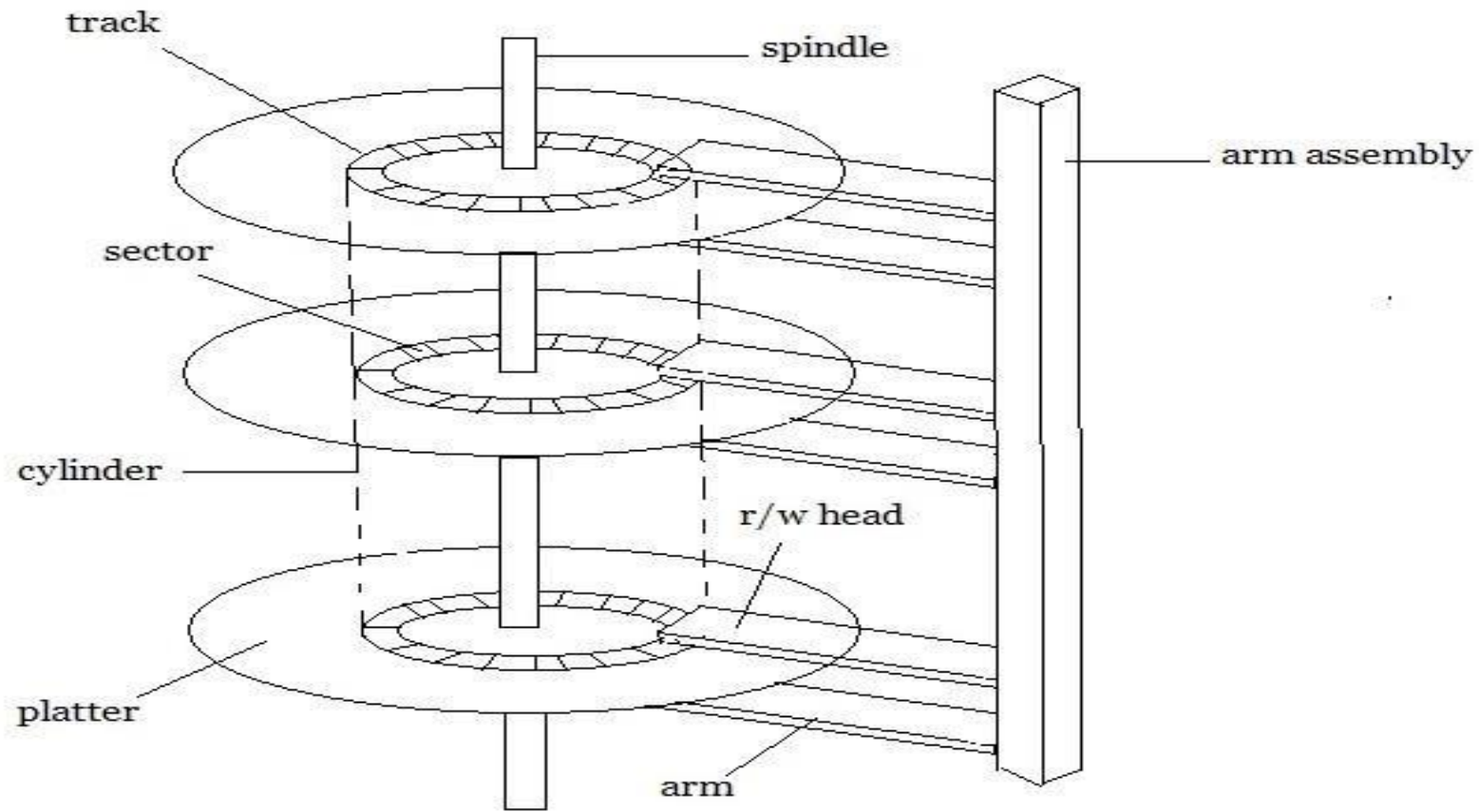


Fig:Disk Structure in OS



Disk Pack Structure

Step wise description of Disk Structure is given below

Disk surface divided into tracks

A read/write head positioned just above the disk surface

Information stored by magnetic recording on the track under read/write head

Fixed head disk

Moving head disk

Designed for large amount of storage

Primary design consideration cost, size, and speed

Hardware for disk system

Disk drive, Device motor, Read/write head, Associated logic

Disk controller

Determines the logical interaction with the computer

Can service more than one drive (overlapped seeks)

Cylinder

The same numbered tracks on all the disk surfaces

Each track contains between 8 to 32 sectors

Sector

Smallest unit of information that can be read from/written into disk

Range from 32 bytes to 4096 bytes Disk Performance Parameters

Some important parameters used to measure the performance of hard disk are as follow Seek time

Seek Time is time required by read/write head to move to requested track

Includes the initial startup time and the time required to traverse the tracks to be crossed once the access arm is up to speed.

Latency or rotational delay

Time required for the requested sector to come under the read/write head.

Rotational delay is generally the half of the time taken in one rotation. Data Transfer Rate

Data transfer rate is define as the amount of data transfer in per unit time for example 30 MB/Sec. Data

Transfer Time

Data Transfer time is the total time taken to transfer a specific amount of data from the disk. Data Transfer

time depends on the data transfer rate of the disk.

Average Access Time

Average access time is calculated as

Average Access Time = Seek Time + Rotational Latency + Data Transfer Time

DISK SCHEDULING AND MANAGEMENT

Disk scheduling and management are essential components of a computer's operating system that handle the organization and access of data on a disk. Disk scheduling algorithms determine the order in which the read/write head of the disk moves to access data, which impacts the efficiency and speed of accessing data. Some commonly used disk scheduling algorithms include First-Come-First-Serve, Shortest Seek Time First, and SCAN. Disk management, on the other hand, involves tasks such as disk partitioning, formatting, and file system creation. It ensures that the disk is properly utilized, optimized for performance, and maintained to prevent data loss or corruption. Efficient disk scheduling and management are crucial for ensuring the smooth and effective operation of a computer system.

Types of Disk Scheduling Algorithms

First-Come, First-Served (FCFS) – The FCFS algorithm is the simplest disk scheduling algorithm, in which the requests are processed in the order they arrive. It is a nonpreemptive algorithm that does not take into account the distance between the current request and the next request.

Shortest Seek Time First (SSTF) – The SSTF algorithm processes the request with the shortest seek time, i.e., the request that requires the least amount of movement by the disk's head. This algorithm can minimize the head movement, but it may result in starvation of some requests.

SCAN – The SCAN algorithm processes the requests in a specific direction, either from the innermost track to the outermost track or vice versa. After reaching the end of the disk, the head reverses direction and processes the requests in the opposite direction. This algorithm provides a balance between the number of requests processed and the time each request waits.

C-SCAN – The C-SCAN algorithm is similar to the SCAN algorithm, but the head moves only in one direction and processes requests only in that direction. When it reaches the end of the disk, it immediately returns to the beginning of the disk and starts processing requests again. This algorithm can eliminate the possibility of starvation.

NOTES: REFER TO THE PROBLEMS THAT I HAVE COVERED IN THE LECTURE

5.2 System Security and Protection

Many companies possess valuable information they want to guard closely. This information can be technical (e.g., a new chip design or software), commercial (e.g., studies of the competition or marketing plans), financial (e.g., plans for a stock offering) legal (e.g., documents about a potential merger or takeover), among many other possibilities. Frequently this information is protected by having a uniformed guard at the building entrance who checks to see that everyone entering the building is wearing a proper badge. In addition, many offices may be locked and some file cabinets may be locked as well to ensure that only authorized people have access to the information.

Home computers increasingly have valuable data on them, too. Many people keep their financial information, including tax returns and credit card numbers, on their computer. Love letters have gone digital. And hard disks these days are full of important photos, videos, and movies. As more and more of this information is stored in computer systems, the need to protect it is becoming increasingly important. Guarding this information against unauthorized usage is therefore a major concern of all operating systems.

Unfortunately, it is also becoming increasingly difficult due to the widespread acceptance of system bloat (and the accompanying bugs) as a normal phenomenon. In the following sections we will look at a variety of issues concerned with security and protection, some of which have analogies to real-world protection of information on paper, but some of which are unique to computer systems. In this chapter we will examine computer security as it applies to operating systems.

Security refers to providing a protection system to computer system resources such as CPU, memory, disk, software programs and most importantly data/information stored in the computer system. If a computer program is run by unauthorized user then he/she may cause severe damage to computer or data stored in it. So a computer system must be protected against unauthorized access, malicious access to system memory, viruses, worms etc.

- Impossible to have absolute security, but make cost to perpetrator sufficiently high to

deter most intruders.

- Security must occur at four levels to be effective:
 - **Physical**
 - ▶ Data centers, servers, connected terminals
 - **Human**
 - ▶ Avoid **social engineering, phishing, dumpster diving**
 - **Operating System**
 - ▶ Protection mechanisms, debugging
 - **Network**
 - ▶ Intercepted communications, interruption, DOS
- Security is as weak as the weakest link in the chain But can too much security be a problem?

From a security perspective, computer systems have four general goals, with corresponding threats to them, as listed. The first, **data confidentiality**, is concerned with having

secret data remain secret. More specifically, if the owner of some data has decided that these data are only to be made available to certain people and no others, the system should guarantee that release of the data to unauthorized people never occurs. As an absolute minimum, the owner should be able to specify who can see what, and the system should enforce these specifications, which ideally should be per file.

The second goal, **data integrity**, means that unauthorized users should not be able to modify any data without the owner's permission. Data modification in this context includes not only changing the data, but also removing data and adding false data. If a system cannot guarantee that data deposited in it remain unchanged until the owner decides to change them, it is not worth much as an information system.

The third goal, **system availability**, means that nobody can disturb the system to make it unusable. Such **denials of service attacks** are increasingly common. For example, if a computer is an Internet server, sending a flood of requests to it may cripple it by eating up all of its CPU time just examining and discarding incoming requests. If it takes, say, 100 uses to process an

incoming request to read a Web page, then anyone who manages to send 10,000 requests/sec can wipe it out. Reasonable models and technology for dealing with attacks on confidentiality and integrity are available; foiling these denial-of-services attacks is much harder.

Finally, a new threat has arisen in recent years. **Outsiders** can sometimes take command of people's home computers (using viruses and other means) and turn them into zombies, willing to do the outsider's bidding at a moment's notice. Often zombies are used to send spam so that the mastermind behind the spam attack cannot be traced. In a certain sense, another threat also exists, but it is more of a threat to society than to an individual users. There are folks out there who bear a grudge against some particular country or (ethnic) group or who are just angry at the world in general and want to destroy as much infrastructure as they can without too much regard to the nature of the damage or who the specific victims are. Usually such people feel that attacking their enemies' computers is a good thing, but the attacks themselves may not be well focused.

Protection:

Protection refers to a mechanism for controlling the access of programs, processes, or users to the resources defined by a computer system.

- ♣ Protection ensures that the resources of the computer are used in a consistent way.

- ♣ It ensures that each object is accessed correctly and only by those processes that are allowed to do so.

Goals of Protection:

- ♣ As computer systems have become more sophisticated and pervasive in their applications, the need to protect their integrity has also grown.

- ♣ We need to provide protection for several reasons. The most obvious is the need to prevent the mischievous, intentional violation of an access restriction by user.

♣ An unprotected resource cannot defend against use (or misuse) by an unauthorized or incompetent user. A protection-oriented system provides means to distinguish between authorized and unauthorized usage.

♣ The role of protection in a computer system is to provide a mechanism for the enforcement of the policies governing resource use. These policies can be established in a variety of ways. Some are fixed in the design of the system, while others are formulated by the management of a system. Still others are defined by the individual users to protect their own files and programs. A protection system must have the flexibility to enforce a variety of policies.

Authentication:

Authentication refers to identifying the each user of the system and associating the executing programs with those users. It is the responsibility of the Operating System to create a protection system which ensures that a user who is running a particular program is authentic. Operating Systems generally identifies/authenticates users using following three ways:

- **Username / Password** - User need to enter a registered username and password with Operating system to login into the system.
- **User card/key** - User need to punch card in card slot, or enter key generated by key generator in option provided by operating system to login into the system.
- **User attribute - fingerprint/ eye retina pattern/ signature** - User need to pass his/her attribute via designated input device used by operating system to login into the system.

THANK YOU !

